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CLAMP ASSEMBLY FOR FALL PROTECTION REINFORCEMENT POST

Abstract

A clamp assembly for a fall protection reinforcement post includes a clamp body at least partially defining a slot configured to receive a guardrail therethrough; a post fastener engaging the clamp body and configured to secure the clamp assembly to the fall protection reinforcement post; and a clamp spring biasing the clamp assembly to an untightened configuration.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. patent application Ser. No. 18/929,242, filed Oct. 28, 2024, which claims the benefit of U.S. Provisional Patent Application No. 63/598,777, filed Nov. 14, 2023, each of which is hereby specifically incorporated by reference herein in its entirety.

TECHNICAL FIELD
[0002] This disclosure relates to building construction. More specifically, this disclosure relates to a clamp assembly for a reinforcement post of a fall protection system.
BACKGROUND
[0003] Building structures under construction often have elevated surfaces, such as elevated floor slabs, that require guardrails or other safety measures to prevent accidental falls. Guardrails are typically placed at a perimeter of the elevated surface and are secured to structural components of the building structure, such as pillars. However, guardrails that extend long distances between adjacent pillars can lack strength and stability at and around their midpoints.
SUMMARY
[0004] It is to be understood that this summary is not an extensive overview of the disclosure. This summary is exemplary and not restrictive, and it is intended neither to identify key or critical elements of the disclosure nor delineate the scope thereof. The sole purpose of this summary is to explain and exemplify certain concepts of the disclosure as an introduction to the following complete and extensive detailed description.
[0005] Disclosed is clamp assembly for a fall protection reinforcement post, the clamp assembly comprising a clamp body at least partially defining a slot configured to receive a guardrail therethrough; a post fastener engaging the clamp body and configured to secure the clamp assembly to the fall protection reinforcement post; a slot stop engaging the clamp body and traversing a bottom slot end of the slot, the slot stop configured to prevent the guardrail from disengaging the slot at the bottom slot end; and a clamp spring biasing the clamp assembly to an untightened configuration.
[0006] Further disclosed is a reinforcement post assembly comprising a fall protection reinforcement post defining a front post side and a rear post side; and a clamp assembly comprising: a clamp body; a post fastener engaging the clamp body and the reinforcement post to secure the clamp assembly to the reinforcement post; and a clamp spring biasing the clamp assembly to an untightened configuration; wherein a slot is defined between the clamp body and the front post side of the reinforcement post, the slot configured to receive a guardrail therethrough.
[0007] Also disclosed is a clamp assembly for a fall protection reinforcement post, the clamp assembly comprising a clamp body at least partially defining a slot configured to receive a guardrail therethrough; a post fastener engaging the clamp body and configured to secure the clamp assembly to the fall protection reinforcement post; and a clamp spring biasing the clamp assembly to an untightened configuration.
[0008] Various implementations described in the present disclosure may include additional systems, methods, features, and advantages, which may not necessarily be expressly disclosed herein but will be apparent to one of ordinary skill in the art upon examination of the following detailed description and accompanying drawings. It is intended that all such systems, methods, features, and advantages be included within the present disclosure and protected by the accompanying claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The features and components of the following figures are illustrated to emphasize the general principles of the present disclosure. Corresponding features and components throughout the figures may be designated by matching reference characters for the sake of consistency and clarity.

[0010] FIG. **1** is a front perspective view of a fall protection system, in accordance with one aspect of the present disclosure.

[0011] FIG. **2** is a front perspective view of a reinforcement post assembly for a fall protection system, in accordance with another aspect of the present disclosure.

[0012] FIG. **3** is a rear perspective view of a guardrail clamp assembly of the reinforcement post assembly of FIG. **2**.

[0013] FIG. **4** is a front perspective view of the guardrail clamp assembly of FIG. **3**.

[0014] FIG. **5** is a left side view of the guardrail clamp assembly of FIG. **3**.

[0015] FIG. **6** is a cross-sectional view of the guardrail clamp assembly of FIG. **3**, taken along line **6-6** in FIG. **5**.

[0016] FIG. **7** is a rear perspective view of a clamp body of the guardrail clamp assembly of FIG. **3**.

[0017] FIG. **8** is a front perspective view of a clamp spring of the guardrail clamp assembly of FIG. **3**.

[0018] FIG. **9** is a front perspective view of a top end of a reinforcement post of the reinforcement post assembly of FIG. **2**.

[0019] FIG. **10** is a rear perspective view of the top end of the reinforcement post of FIG. **9**.

[0020] FIG. **11** is a cross-sectional view of the guardrail clamp assembly of FIG. **3**, taken along line **6-6** in FIG. **5**, wherein the guardrail clamp assembly is in an untightened configuration.

[0021] FIG. **12** is a cross-sectional view of the guardrail clamp assembly of FIG. **3**, taken along line **6-6** in FIG. **5**, showing a first step in tightening the guardrail clamp assembly.

[0022] FIG. **13** is a cross-sectional view of the guardrail clamp assembly of FIG. **3**, taken along line **6-6** in FIG. **5**, showing a second and final step in tightening the guardrail clamp assembly to the tightened configuration.

[0023] FIG. **14** is a cross-sectional view of the guardrail clamp assembly of FIG. **3**, taken along line **6-6** in FIG. **5**, showing a first step in reconfiguring a slot stop from an unblocked position to a blocked position.

[0024] FIG. **15** is a left side view of a bottom end of the reinforcement post of FIG. **9**.

[0025] FIG. **16** is a cross-sectional view of the guardrail clamp assembly according to another aspect of the present disclosure, taken along a line similar to line **6-6** in FIG. **5**.

DETAILED DESCRIPTION

[0026] The present disclosure can be understood more readily by reference to the following detailed description, examples, drawings, and claims, and the previous and following description. However, before the present devices, systems, and/or methods are disclosed and described, it is to be understood that this disclosure is not limited to the specific devices, systems, and/or methods disclosed unless otherwise specified, and, as such, can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

[0027] The following description is provided as an enabling teaching of the present devices, systems, and/or methods in its best, currently known aspect. To this end, those skilled in the relevant art will recognize and appreciate that many changes can be made to the various aspects of the present devices, systems, and/or methods described herein, while still obtaining the beneficial results of the present disclosure. It will also be apparent that some of the desired benefits of the present disclosure can be obtained by selecting some of the features of the present disclosure without utilizing other features. Accordingly, those who work in the art will recognize that many modifications and adaptations to the present disclosure are possible and can even be desirable in certain circumstances and are a part of the present disclosure. Thus, the following description is provided as illustrative of the principles of the present disclosure and not in limitation thereof.

[0028] As used throughout, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “an element” can include two or more such elements unless the context indicates otherwise.

[0029] Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another aspect includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

[0030] For purposes of the current disclosure, a material property or dimension measuring about X or substantially X on a particular measurement scale measures within a range between X plus an industry-standard upper tolerance for the specified measurement and X minus an industry-standard lower tolerance for the specified measurement. Because tolerances can vary between different materials, processes and between different models, the tolerance for a particular measurement of a particular component can fall within a range of tolerances.

[0031] As used herein, the terms “optional” or “optionally” mean that the subsequently described event or circumstance can or cannot occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.

[0032] The word “or” as used herein means any one member of a particular list and also includes any combination of members of that list. Further, one should note that conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain aspects include, while other aspects do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular aspects or that one or more particular aspects necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular aspect.

[0033] Disclosed are components that can be used to perform the disclosed methods and systems. These and other components are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these components are disclosed that while specific reference of each various individual and collective combinations and permutations of these may not be explicitly disclosed, each is specifically contemplated and described herein, for all methods and systems. This applies to all aspects of this application including, but not limited to, steps in disclosed methods. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific aspect or combination of aspects of the disclosed methods.

[0034] Disclosed is a guardrail clamp assembly and associated methods, systems, devices, and various apparatus. Example aspects of the guardrail clamp assembly can comprise a clamp body, a post fastener, a slot stop, and a clamp spring. It would be understood by one of skill in the art that the guardrail clamp assembly is described in but a few exemplary embodiments among many. No particular terminology or description should be considered limiting on the disclosure or the scope of any claims issuing therefrom.

[0035] FIG. **1** is a front perspective view of a fall protection system **100**, in accordance with one aspect of the present disclosure. The fall protection system **100** can be secured to a structural component of a structure, such as a building structure **140** under construction, to prevent accidental falls from an upper floor surface **146** of an elevated slab **145** or other elevated surface of the building structure **140**. The structural component can be a substantially vertical pillar **150**, for example and without limitation. In the present aspect, the building structure **140** can comprise the elevated slab **145** and at least one of the substantially vertical pillars **150** extending upward therefrom, and the fall protection system **100** can be coupled to the at least one pillar **150**. In the present aspect, the at least one pillar **150** can comprise a first pillar **150***a* and a second pillar **150***b*, and the fall protection system **100** can be coupled to and can extend between the first and second pillars **150***a,b*. In other aspects, the building structure **140** can comprise more or fewer pillars **150**. For example, the building structure **140** may comprise additional pillars **150** and the fall protection system **100** can be coupled to and can extend between the additional pillars **150**. Example aspects of such a fall protection system **100** are further disclosed in U.S. Pat. No. 11,753,836, issued on Sep. 12, 2023, and U.S. Pat. No. 11,795,713, issued on Oct. 24, 2023, each of which is hereby specifically incorporated by reference herein in its entirety.

[0036] Example aspects of the fall protection system **100** can comprise a guardrail support frame **108**. The guardrail support frame **108** can comprise at least one substantially vertical guardrail post **110**. Each of the guardrail posts **110** can be secured to a corresponding one of the vertical pillars **150**. For example, in the present aspect, the guardrail posts **110** can comprise at least a first guardrail post **110***a* secured to the first pillar **150***a* and a second guardrail post **110***b* secured to the second pillar **150***b*. According to example aspects, each of the guardrail posts **110** can comprise a metal material, such as steel. In some aspects, the guardrail posts **110** can be formed as extruded steel posts. In other aspects, the guardrail posts **110** can comprise any other suitable material known in the art, including but not limited to other metals, and/or can be manufactured using any other desired method. In other aspects, the guardrail posts **110** can comprise a non-metal material. Moreover, in other aspects, the fall protection system **100** can comprise more or fewer guardrail posts **110**. For example, the fall protection system **100** may comprise additional guardrail posts **110**, each of which can be secured to an additional pillar **150**.

[0037] The fall protection system **100** can further comprising one or more lateral safety guardrails **125** supported by the guardrail support frame **108**. In the present aspect, the lateral safety guardrails **125** can extend laterally between the first and second guardrail posts **110***a,b*, and thus, can extend between the first and second pillars **150***a,b*. The lateral safety guardrails **125** can be oriented about horizontally, as shown in the present aspect, or can be oriented at an acute angle relative to horizontal. In the present aspect, each of the lateral safety guardrails **125** can be formed as a flexible guardrail strap **125**. In other aspects, the lateral safety guardrails **125** can be semi-rigid or rigid. Furthermore, in the present aspect, the guardrail straps **125** can comprise an upper guardrail strap **125***a*, a middle guardrail strap **125***b*, and a lower guardrail strap **125***c*. Other aspects of the fall protection system **100** can comprise more or fewer of the guardrail straps **125** extending between the first and second guardrail posts **110***a,b*. Example aspects of the guardrail straps **125** can comprise a plastic material, such as, for example, nylon webbing. In other aspects, the guardrail straps **125** can comprise any other suitable material known in the art. In some aspects, the guardrail straps **125** can be inelastic or semi-inelastic, while in other aspects, the guardrail straps **125** can be elastic. Additionally, in example aspects, a length of each guardrail strap **125** can be sized to accommodate varying distances between the first and second guardrail posts **110***a,b*.

[0038] Example aspects of the guardrail support frame **108** can further comprise one or more substantially vertical reinforcement post assemblies **200** (shown in FIG. **2**) arranged between the adjacent guardrail posts **110** (e.g., the first and second guardrail posts **110***a,b*) and coupled to each of the guardrail straps **125**. The reinforcement post assemblies **200** can be freestanding (i.e., not attached to the building structure **140**) and can provide reinforcement to the guardrail straps **125** between the adjacent guardrail posts **110**, which can be beneficial in building structures **140** wherein a significant distance is defined between adjacent pillars **150**. Each of the reinforcement post assemblies **200** can comprise a reinforcement post **210** (shown in FIG. **2**). In example aspects, the reinforcement post **210** can comprise a metal material, such as steel. In some aspects, the reinforcement posts **210** can be formed as extruded steel posts, similar to the guardrail posts **110**. In other aspects, the reinforcement posts **210** can comprise any other suitable material known in the art, including but not limited to other metals, and/or can be manufactured using any other desired method.

[0039] According to example aspects, each of the guardrail straps **125** can be mounted to the first guardrail post **110***a* by a corresponding guardrail attachment assembly **120**. The guardrail attachment assemblies **120** can be vertically spaced along a length of the first guardrail post **110***a*, as shown. In example aspects, each of the guardrail straps **125** can be rolled up on the corresponding guardrail attachment assembly **120**. To extend each of the guardrail straps **125** from the first guardrail post **110***a* to the second guardrail post **110***b*, the guardrail strap **125** can be unrolled or partially unrolled by pulling a free end **126** of the guardrail strap **125** towards the second guardrail post **110***b*. The free end **126** can be releasably secured to the second guardrail post **110***b* and the guardrail strap **125** can be held taut between the first and second guardrail posts **110***a*,*b*, as described in further detail below. With the guardrail straps **125** extended between the first and second guardrail posts **110***a*,*b*, the fall protection system **100** can define a substantially upright boundary **105** between the first and second pillars **150***a*,*b*, as shown.

[0040] As previously described, in some aspects, the fall protection system **100** can extend between additional pillars **150**. As shown, in the present aspect, the fall protection system **100** can comprise a plurality of additional guardrail straps **125**, each mounted to the second guardrail post **110***b* by an additional one of the guardrail attachment assemblies **120**. Each of the additional guardrail straps **125** can be configured to extend between the second pillar **150***b* and an adjacent third pillar (not shown) of the building structure **140**.

[0041] Example aspects of the fall protection system **100** can further comprise one or more pillar attachment assemblies **130**. Each of the pillar attachment assemblies **130** can be configured to couple a corresponding one of the guardrail posts **110** to the corresponding vertical pillar **150**. Securing the upright boundary **105** to the vertical pillars **150** can prevent movement thereof and ensure safe and proper functioning of the fall protection system **100**. For example, the pillar attachment assemblies **130** can comprise a first pillar attachment assembly **130***a* coupling the first guardrail post **110***a* to the first pillar **150***a* and a second pillar attachment assembly **130***b* coupling the second guardrail post **110***b* to the second pillar **150***b*. As shown, each pillar attachment assembly **130** can comprise a pillar attachment device **132** and a pillar attachment strap **135** that can be rolled up on the pillar attachment device **132**.

[0042] Each pillar attachment device **132** can be mounted to the corresponding guardrail post **110**, and the corresponding pillar attachment strap **135** can extend from the pillar attachment device **132** and wrap fully around the corresponding pillar **150**. To wrap the pillar attachment strap **135** around the pillar **150**, the pillar attachment strap **135** can be unrolled or partially unrolled by pulling a free end thereof around the pillar **150**. The free end can then be releasably coupled to the corresponding guardrail post **110** to secure the pillar attachment strap **135** tautly around the pillar **150**. In the present aspect, only one of the pillar attachment assemblies **130** secures each of the first and second guardrail posts **110***a*,*b* to the corresponding first and second pillars **150***a*,*b*. However, in other aspects, the fall protection system **100** can comprise additional pillar attachment assemblies **130** for securing each of the first and second guardrail posts **110***a*,*b* to the first and second pillars **150***a*,*b*. According to example aspects, a length of each pillar attachment strap **135** can be sized to accommodate pillars **150** of varying thicknesses.

[0043] FIG. **2** illustrates one of the substantially vertical reinforcement post assemblies **200** arranged at or around a midpoint between the adjacent guardrail posts **110** (shown in FIG. **1**) and coupled to each of the guardrail straps **125** (e.g., the upper guardrail strap **125***a*, the middle guardrail strap **125***b*, and the lower guardrail strap **125***c*). In other aspects, the reinforcement assembly **200** can be coupled to more or fewer guardrail straps **125**. In other aspects, additional reinforcement post assemblies **200** can be arranged between the adjacent guardrail posts **110** and coupled to the guardrail straps **125**. The reinforcement post **210** of the reinforcement post assembly

200 can define an upper post end **212**, a lower post end **214** opposite the upper post end **212**, and a middle post region **216** therebetween. The lower post end **214** can generally confront or rest on the upper floor surface **146** (shown in FIG. **1**) of the elevated slab **145** (shown in FIG. **1**), and the reinforcement post **210** can extend substantially vertically upward therefrom, as shown. In some aspects, a post end cap **910** (shown in FIG. **9**) can be coupled to the reinforcement post **210** at the upper post end **212** thereof to cover an upper post opening **610** (shown in FIG. **6**). The upper post opening **610** can allow access into a hollow center **612** (shown in FIG. **6**) of the reinforcement post **210**. The post end cap **910** can comprise a rubber material in some aspects; in other aspects, the post end cap **910** can comprise plastic, metal, or any other suitable material. The reinforcement post **210** can further define a front post side **218** and a rear post side **320** (shown in FIG. **3**) opposite the front post side **218**.

[0044] According to example aspects, a strap engagement slot **222** can extend substantially vertically into the reinforcement post **210** at the lower post end **214**. The lower guardrail strap **125***c* can extend laterally through the strap engagement slot **222** to secure the lower guardrail strap **125***c* to the reinforcement post assembly **200**. The strap engagement slot **222** is also shown in FIG. **15** and described in further detail below. Additionally, the reinforcement post assembly **200** can comprise one or more guardrail clamp assemblies **230** attached to the reinforcement post **210**, as shown. Each of the guardrail clamp assemblies **230** can be coupled to a corresponding one of the guardrail straps **125**. In the present aspect, an upper clamp assembly **230***a* of the guardrail clamp assemblies **230** can couple the upper guardrail strap **125***a* to the reinforcement post **210** proximate to the upper post end **212**, and a middle clamp assembly **230***b* of the guardrail clamp assemblies **230** can couple the middle guardrail strap **125***b* to the reinforcement post **210** at the middle post region **216**. In other aspects, a lower one of the guardrail clamp assemblies **230** can be attached to the reinforcement post **210** at the lower post end **214** thereof to couple the lower guardrail strap **125***c* thereto. Such a lower one of the guardrail clamp assemblies **230** can be provided instead of or in addition to the strap engagement slot **222**.

[0045] In example aspects, a clamp end cap **224** can be coupled to the upper clamp assembly **230***a*. The clamp end cap **224** can comprise a rubber material in some aspects; in other aspects, the clamp end cap **224** can comprise plastic, metal, or any other suitable material. The middle clamp assembly **230***b* can be substantially the same as the upper clamp assembly **230***a*, without the clamp end cap **224**. According to example aspects, the upper clamp assembly **230***a* can be slipped over reinforcement post **210** at either the upper post end **212** or the opposing lower post end **214** and can be slid along the reinforcement post **210** to position the upper clamp assembly **230***a* at or proximate to the upper post end **212**. Similarly, the middle clamp assembly **230***b* can be slipped over the reinforcement post **210** at either the upper post end **212** or the opposing lower post end **214** and can be slid along the reinforcement post **210** to position the middle clamp assembly **230***b* at the middle post region **216**.

[0046] Additionally, according to example aspects, the reinforcement post assembly **200** can comprise a fall arrest bracket **240** coupled to the reinforcement post **210**. As described in further detail below, a user safety harness can be attached to the fall arrest bracket **240**. In the present aspect, the fall arrest bracket **240** can be coupled to the reinforcement post **210** between the upper clamp assembly **230***a* and the middle clamp assembly **230***b*. In other aspects, the fall arrest bracket **240** can be positioned at any suitable location along the reinforcement post **210**. In other aspects, the reinforcement post assembly **200** can comprise additional fall arrest brackets **240** or may not comprise the fall arrest bracket **240**. The reinforcement post assembly **200** may comprise any other suitable fall arrest device(s) attached thereto.

[0047] FIGS. **3**, **4**, and **5**, illustrate a rear perspective view, a front perspective view, and a left-side view, respectively, of the upper clamp assembly **230***a* and the fall arrest bracket **240** attached to the reinforcement post **210**. According to example aspects, the fall arrest bracket **240** can be coupled to the reinforcement post **210** by a pair of bracket fasteners **302**. In other aspects, more or fewer of the bracket fasteners **302** can couple the fall arrest bracket **240** to the reinforcement post **210**. The bracket fasteners **302** can be bolts and nuts (as shown), screws, rivets, welding, or any other suitable fastener known in the art. As shown, the fall arrest bracket **240** can define one or more safety attachment openings **304** formed therethrough for securing the user safety harness to the fall arrest bracket **240**. The fall arrest bracket **240** can comprise a metal material, composite material, plastic material, or any material of suitable strength. In some aspects, the fall arrest bracket **240** can be monolithically formed (i.e., formed a singular component that constitutes a single material without joints or seams) from a single piece of material and bent to the desired shape. Other aspects of the fall arrest bracket **240** may not be monolithically formed.

[0048] Each of the guardrail clamp assemblies **230** can comprise a clamp body **330**, a post fastener **332** that can secure the clamp body **330** to the reinforcement post **210**, and a clamp spring **340** that can bias the guardrail clamp assembly **230** to an untightened configuration (as shown in FIGS. **3**, **4**, and **5**). A clamp slot **344** can be defined generally between a front clamp wall **350** of the clamp body **330** and the front post side **218** of the reinforcement post **210**. The corresponding guardrail strap **125** (shown in FIG. **1**) can extend laterally through the clamp slot **344**. In the untightened configuration, a bottom slot end **346** of the clamp slot **344** can be open to allow the guardrail strap **125** to be inserted vertically into the slot. However, the clamp slot **344** can be closed at an opposite upper slot end **348** thereof to retain the guardrail strap **125** within the clamp slot **344** at the upper slot end **348**. As shown, the clamp body **330** of the guardrail clamp assembly **230** can define an upper slot boundary **354** that can close the clamp slot **344** at the upper slot end **348**. Each of the guardrail clamp assemblies **230** can further define a primary slot stop **360** mounted to the clamp body **330** that can adjustably engage the reinforcement post **210** to close the bottom slot end **346** in the tightened configuration and retain the guardrail strap **125** within the clamp slot **344**.

[0049] The post fastener **332** can be untightened and tightened to adjust a width of the clamp slot **344** to reconfigure the guardrail clamp assembly **230** between the untightened configuration and a tightened configuration (shown in FIGS. **13** and **14**). In the untightened configuration, the guardrail strap **125** can be somewhat loosely received within the clamp slot **344**, and in the tightened configuration, the guardrail strap **125** can be clamped tightly between the clamp body **330** and the reinforcement post **210**. In the present aspect, the post fastener **332** can be a threaded bolt fastener **334**, which can extend through a rear clamp wall **352** of the clamp body **330**, opposite the front clamp wall **350**, and into the reinforcement post **210** at the rear post side **320**. The threaded bolt fastener **334** can comprise a threaded shaft **336** that can engage a corresponding threaded fastener bore **622** (shown in FIG. **6**) of the guardrail clamp assembly **230**. The threaded bolt fastener **334** can further comprise an adjustment knob **338** coupled to the threaded shaft **336**. The adjustment knob **338** can be rotated to tighten or untighten the threaded shaft **336** relative to the clamp body **330**.

[0050] In example aspects, the clamp spring **340** can be arranged between the front post side **218** of the reinforcement post **210** and the front clamp wall **350** of the clamp body **330**. The clamp spring **340** can bias the front clamp wall **350** away from the front post side **218** to the untightened configuration. In the untightened configuration, the bottom slot end **346** can be open and the guardrail strap **125** can be loosely received within the clamp slot **344**. As the post fastener **332** is tightened, the spring force of the clamp spring **340** can be overcome and the front clamp wall **350** can be drawn towards the front post side **218**, as described in further detail below. In the present aspect, the clamp spring **340** can be a V-shaped button spring clip **342**. The button spring clip **342** can be formed from stainless steel in example aspects; however, in other aspects, the button spring clip **342** can be formed from any other suitably resilient material. In other aspects, the clamp spring **340** can be any other suitable spring known in the art.

[0051] The primary slot stop **360** can be arranged in an unblocked position, wherein the bottom slot end **346** of the clamp slot **344** is unblocked by the primary slot stop **360**, and a blocked position, wherein the bottom slot end **346** of the clamp slot **344** is blocked by the primary slot stop **360**.

According to example aspects, the primary slot stop **360** can be substantially similar to the post fastener **332**. That is, in example aspects, the primary slot stop **360** can be another one of the threaded bolt fasteners **334**, which can extend through the front clamp wall **350** of the clamp body **330**. The threaded bolt fastener **334** can comprise the threaded shaft **336** and the adjustment knob **338**. The threaded shaft **336** of the primary slot stop **360** can engage a corresponding threaded stop bore **632** (shown in FIG. **6**) of the guardrail clamp assembly. The adjustment knob **338** can be rotated to untighten or tighten the threaded shaft **336** of the primary slot stop **360** relative to the clamp body **330** between the unblocked and blocked positions, respectively. In the unblocked position, a distal stop end **362** of the threaded shaft **336** of the primary slot stop **360** can be spaced from the front post side **218** of the reinforcement post **210**, and in the blocked position, the distal stop end **362** of the threaded shaft **336** can traverse the clamp slot **344** and can extend into the reinforcement post **210** at the front post side **218**. In other aspects, the primary slot stop **360** can be formed as any other suitable device that can be selectively reconfigured to block and unblock the clamp slot **344** at or near the bottom slot end **346**.

[0052] In example aspects, a first clamp sidewall **370** and a second clamp sidewall **670** (shown in FIG. **6**) can extend between the front clamp wall **350** and the rear clamp wall **352**. Each of the first clamp sidewall **370** and the second clamp sidewall **670** can define a substantially V-shaped notch **372**. A forward edge **374** of each V-shaped notch **372** can at least partially define the clamp slot **344**. According to example aspects, a plurality of gripping teeth **376** can be formed along the forward edge **374** of each V-shaped notch **372**. The gripping teeth **376** can be configured to grip and retain the guardrail strap **125** in place when the guardrail strap **125** is clamped between the guardrail clamp assembly and the reinforcement post **210** in the tightened configuration. The upper slot boundary **354** of the clamp body **330** can be defined by the notch vertex **378** of each V-shaped notch **372**. Other aspects of the guardrail clamp assembly **230** and/or the reinforcement post **210** can also or alternatively define any other suitable gripping device for aiding in gripping and retaining the guardrail strap **125** in place in the tightened configuration, such as, for example and without limitations, textured portions, non-slip pads (such as rubber pads), projections of any suitable shape and size, and the like.

[0053] Additionally, in some aspects, each of the first clamp sidewall **370** and the second clamp sidewall **670** can define a secondary slot stop **380** at a bottom wall end **356** thereof. Each of the secondary slot stops **380** can be formed as a stop hook **382**. The stop hook **382** can extend towards the rear clamp wall **352**, at least partially traversing the clamp slot **344** in the untightened configuration. In the tightened configuration, the stop hook **382** of the first clamp sidewall **370** can be positioned at a first post side **312** of the reinforcement post **210**, and the stop hook **382** of the second clamp sidewall **670** can be positioned at an opposite second post side **314** of the reinforcement post **210**. Thus, in the tightened configuration, the stop hooks **382** can further block the bottom slot end **346** of the clamp slot **344** to aid in retaining the guardrail strap **125** within the clamp slot **344**.

[0054] FIG. **6** illustrates a cross-sectional view of the upper clamp assembly **230***a* coupled to the reinforcement post **210**, taken along line **6-6** in FIG. **5**. As shown, the clamp body **330** of the guardrail clamp assembly **230** can be mounted to the reinforcement post **210** by the post fastener **332**. A fastener rivet nut **620**, or other suitable nut or fastener defining the threaded fastener bore **622**, can be mounted within a rear clamp opening **624** formed through the rear clamp wall **352** of the clamp body **330**. The fastener rivet nut **620** can define the threaded fastener bore **622**, and the threaded shaft **336** of the post fastener **332** can engage the threaded fastener bore **622**. The threaded shaft **336** of the post fastener **332** can further extend through a rear post opening **626** formed through the rear post side **320** of the reinforcement post **210**.

[0055] Similarly, a stop rivet nut **630**, or other suitable nut or fastener defining the threaded stop bore **632**, can be mounted within a front clamp opening **634** formed through the front clamp wall **350** of the clamp body **330**, proximate to the bottom slot end **346** of the clamp slot **344**. The stop rivet nut **630** can define the threaded stop bore **632**, and the threaded shaft **336** of the primary slot stop **360** can engage the threaded stop bore **632**. The distal stop end **362** of the threaded shaft **336** of the primary slot stop **360** can be spaced from the front post side **218** in the unblocked position, as shown. In the blocked position, the distal stop end **362** of the primary slot stop **360** can extend transversely across the clamp slot **344** and through a front post opening **636** formed through the front post side **218** of the reinforcement post **210**. In some aspects, a guide rivet nut **640** defining a threaded guide bore **642**, or other suitable nut or fastener defining the threaded guide bore **642**, can be mounted within the front post opening **636**, and the threaded shaft **336** of the primary slot stop **360** can engage the threaded guide bore **642** in the blocked position. Other aspects, however, such as the aspect shown in FIG. **16**, do not comprise the guide rivet nut **640**.

[0056] The clamp spring **340** can be the button spring clip **342** in the present aspect, which can be arranged within the clamp slot **344** between the front post side **218** of the reinforcement post **210** and the front clamp wall **350** of the clamp body **330**. The button spring clip **342** can define a substantially V-shaped resilient spring portion **650** and a button tab **652** extending from the substantially V-shaped resilient spring portion **650**. The button spring clip **342** can be inverted such that the V-shaped resilient spring portion **650** can be arranged upside-down, as shown. The V-shaped resilient spring portion **650** can define a first leg **654** confronting an outer post surface **614** of the front post side **218** of the reinforcement post **210** and a second leg **656** confronting an inner clamp wall surface **680** of the front clamp wall **350**. The first leg **654** can define a first end **658** of the V-shaped resilient spring portion **650**, and the second leg **656** can define a second end **660** of the V-shaped resilient spring portion **650**. A spring vertex **662** of the V-shaped resilient spring portion **650** can be oriented vertically above the first end **658** and the second end **660**. The button tab **652** can extend outward from the first leg **654** proximate to the first end **658** and can engage a button opening **616** formed through the front post side **218**. In other aspects, the clamp spring **340** can be arranged between the rear clamp wall **352** and the rear post side **320**.

[0057] FIG. **7** illustrates an example aspect of the clamp body **330**. Example aspects of the clamp body **330** can comprise a metal material, such as laser-cut steel, a plastic material, a composite material, or the like. The clamp body **330** can define the front clamp wall **350**, the rear clamp wall **352**, the first clamp sidewall **370**, and the second clamp sidewall **670**. The rear clamp opening **624** can be formed through the rear clamp wall **352**, and the front clamp opening **634** can be formed through the front clamp wall **350**. In example aspects, each of the rear clamp opening **624** and the front clamp opening **634** can be substantially hexagonal in shape. In other aspects, the rear clamp opening **624** and/or the front clamp opening **634** can define any other suitable shape. In some aspects, the fastener rivet nut **620** (shown in FIG. **6**) and the stop rivet nut **630** (shown in FIG. **6**) can be held in place within the rear clamp opening **624** and the front clamp opening **634**, respectively, by deforming the fastener rivet nut **620** and the stop rivet nut **630** within the rear clamp opening **624** and the front clamp opening **634**, such as by compressing the fastener rivet nut **620** and the stop riven nut **630** such that portions of the fastener rivet nut **620** and the stop riven nut **630** on either side of the rear clamp opening **624** and the front clamp opening **634**, respectively, can be wider than each opening **624**,**634**.

[0058] Each of the first clamp sidewall **370** and the second clamp side wall can define the corresponding substantially V-shaped notch **372**. Each V-shaped notch **372** can define the forward edge **374**, and the plurality of gripping teeth **376** can be formed along the forward edge **374**. In the present aspect, the forward edge **374** of each V-shaped notch **372** can be oriented substantially vertical, while a rearward edge **710** of each V-shaped notch **372** can angle towards the rear clamp wall **352**. Each of the first clamp sidewall **370** and second clamp sidewall **670** can further define the stop hook **382** arranged at the bottom wall end **356** thereof and extending rearward towards the rear clamp wall **352**. Each of the stop hooks **382** can curve slightly upward in the present aspect, relative to the orientation shown.

[0059] FIG. **8** illustrates an example aspect the clamp spring **340**. As shown, the clamp spring **340** can be the button spring clip **342** in the present aspect. The button spring clip **342** can define the substantially V-shaped resilient spring portion **650** (arranged in an inverted orientation) and the button tab **652**. The V-shaped resilient spring portion **650** can define the first leg **654** and the second leg **656**, which can meet at the spring vertex **662** of the V-shaped resilient spring portion **650**. The first leg **654** can define the first end **658** of the V-shaped resilient spring portion **650**, and the second leg **656** can define the second end **660** of the V-shaped resilient spring portion **650**. The button tab **652** can extend outward from the first leg **654**, at or near the first end **658**. In example aspects, the button tab **652** can define one or more relief notches **810** that can allow the button tab **652** to flex as the button tab **652** is pushed through the button opening **616** (shown in FIG. **6**) of the front post side **218** (shown in FIG. **2**). In the present aspect, a first relief notch **810***a* can be formed at a first button side **820** of the button tab **652** and a second relief notch **810***b* can be formed at an opposite second button side **822** of the button tab **652**. Other aspects can comprise more or fewer relief notches **810**.

[0060] FIGS. **9** and **10** illustrate front and rear perspective views, respectively, of the upper post end **212** of the reinforcement post **210**, along with the post fastener **332** and the primary slot stop **360** aligned with the rear post opening **626** (shown in FIG. **10**) and the front post opening **636**, respectively, of the reinforcement post **210**. Referring to FIG. **9**, the threaded shaft **336** of the primary slot stop **360** can engage the stop rivet nut **630**, which can be mounted to the front clamp wall **350** (shown in FIG. **3**) of the clamp body **330** (shown in FIG. **3**). Additionally, the threaded shaft **336** of the primary slot stop **360** can be aligned with, and in the blocked position can engage, the front post opening **636** formed through the front post side **218** of the reinforcement post **210**. In some aspects, the guide rivet nut **640** can be mounted within the front post opening **636**, and the threaded shaft **336** of the primary slot stop **360** can engage the guide rivet nut **640** in the blocked configuration. Other aspects (such as the aspect shown in FIG. **16**) may not comprise the guide rivet nut **640**. Also shown in FIG. **9** is the button opening **616** formed through the front post side **218** of the reinforcement post **210**, as well as the post end cap **910** coupled to the reinforcement post **210** at the upper post end **212**.

[0061] Referring to FIG. **10**, the threaded shaft **336** of the post fastener **332** can engage the fastener rivet nut **620**, which can be mounted to the rear clamp wall **352** (shown in FIG. **3**) of the clamp body **330** (shown in FIG. **3**). Additionally, the threaded shaft **336** of the post fastener **332** can be aligned with and can engage the rear post opening **626** formed through the rear post side **320** of the reinforcement post **210**. In the tightened configuration, the fastener rivet nut **620** may abut the rear post side **320** or may extend, or partially extend, through the rear post opening **626**.

[0062] FIG. **11** illustrates a cross-sectional view of the upper clamp assembly **230***a* mounted to the reinforcement post **210** at the upper post end **212** and in the untightened configuration. As shown, the distal stop end **362** of the threaded shaft **336** of the primary slot stop **360** can be spaced from the front post side **218** of the reinforcement post **210**, and the bottom slot end **346** of the clamp slot **344** can be open and unblocked. The upper guardrail strap **125***a* can be inserted into the clamp slot **344** through the open bottom slot end **346**. Moreover, the gripping teeth **376** of the first clamp sidewall **370** (shown in FIG. **3**) and the second clamp sidewall **670** can be spaced from the upper guardrail strap **125***a*, such that the upper guardrail strap **125***a* can be somewhat loosely received within the clamp slot **344**. As shown, the clamp spring **340** arranged between the front post side **218** and the front clamp wall **350** can bias the front clamp wall **350** and the gripping teeth **376** forwardly away from the front post side **218** and the upper guardrail strap **125***a* in the untightened configuration. Moreover, in the untightened configuration, a distal fastener end **1110** of the threaded shaft **336** of the post fastener **332** can extend through the rear post side **320** of the reinforcement post **210** to be positioned within the hollow center **612** of the reinforcement post **210**. In other aspects, the distal fastener end **1110** can be arranged external to the hollow center **612** in the untightened configuration. In some aspects, the distal fastener end **1110** can be spaced from the front post side **218** of the reinforcement post **210** in the untightened configuration, as shown, while in other aspects, the distal fastener **1110** may confront the front post side **218** in the untightened configuration.

[0063] FIG. **12** illustrates a first step in tightening the post fastener **332** to configure the upper clamp assembly **230***a* in the tightened configuration. By rotating the adjustment knob **338** of the post fastener **332**, the threaded shaft **336** of the post fastener **332** can be tightened relative to the fastener rivet nut **620** to advance the distal fastener end **1110** of the threaded shaft **336** towards an inner post surface **1220** of the front post side **218**. Once the distal fastener end **1110** has been advanced forward to abut the inner post surface **1220**, the threaded shaft **336** of the post fastener **332** can further be rotated relative to the fastener rivet nut **620** to draw the fastener rivet nut **620** and rear clamp wall **352** towards the adjustment knob **338** of the post fastener **332**. As the rear clamp wall **352** is drawn towards the adjustment knob **338** of the post fastener **332** and away from the rear post side **320**, the front clamp wall **350** can be pressed against the clamp spring **340** and can overcome the spring force to be drawn towards the front post side **218**. In example aspects, the front clamp wall **350** can first be pulled towards the front post side **218** at an upper clamp end **1230** of the clamp body **330** due to the inverted V-shape of the resilient spring portion **650** of the clamp spring **340**.

[0064] FIG. **13** illustrates a second and final step in tightening the post fastener **332** to configure the upper clamp assembly **230***a* in the tightened configuration. As shown, upon even further rotation of the threaded shaft **336** of the post fastener **332** relative to the fastener rivet nut **620**, the spring force of the clamp spring **340** can be overcome and the front clamp wall **350** can be drawn towards the front post side **218** of the reinforcement post **210** at a lower clamp end **1310** of the clamp body **330**. The front clamp wall **350** can be drawn towards the front post side **218** of the reinforcement post **210** until the gripping teeth **376** formed along the forward edge **374** of each of the first clamp sidewall **370** (shown in FIG. **3**) and the second clamp sidewall **670** engage and hold the upper guardrail strap **125***a* against the front post side **218**. The upper clamp assembly **230***a* can thereby be configured in the tightened configuration, and the upper guardrail strap **125***a* can be clamped in place between the clamp body **330** and the reinforcement post **210**.

[0065] In some example aspects, the front clamp wall **350** can be wider than the front post side **218**. When the post fastener **332** is tightened to configure the clamp assembly **230** in the tightened configuration, the forward edge **374** of each of the first clamp sidewall **370** and the second clamp sidewall **670** can be drawn past the front post side **218** to be arranged at the first post side **312** (shown in FIG. **3**) and the second post side **314**, respectively. The gripping teeth **376** can engage side portions of the guardrail strap **125** arranged at either side of the front post side **218** and can push the side portions of the guardrail strap **125** rearward towards the rear post side **320**. The guardrail strap **125** can thereby wrap around the front post side **218** of the reinforcement post **210** and extend partially along the first post side **312** and second post side **314** thereof.

[0066] FIG. **14** illustrates the upper clamp assembly **230***a* in the tightened configuration and the primary slot stop **360** in the blocked position. To configure the primary slot stop **360** in the blocked position, the adjustment knob **338** of the primary slot stop **360** can be rotated to tighten the threaded shaft **336** of the primary slot stop **360** relative to the stop rivet nut **630**. Tightening the threaded shaft **336** of the primary slot stop **360** can advance the distal stop end **362** of the threaded shaft **336** across the clamp slot **344** blocking the clamp slot **344** at the bottom slot end **346**, and through the front post opening **636** of the reinforcement post **210**. In some aspects, the guide rivet nut **640** can be mounted within the front post opening **636**, and the threaded shaft **336** of the primary slot stop **360** can further engage the threaded guide bore **642** of the guide rivet nut **640**. With the primary slot stop **360** in the blocked position, the upper guardrail strap **125***a* can be retained within the clamp slot **344** between the closed upper slot end **348** and the threaded shaft **336** of the primary slot stop **360**. As previously described, the secondary slot stops **380** (shown in FIG. **3**), if present, can further aid in retaining the upper guardrail strap **125***a* within the clamp slot **344**.

[0067] FIG. **15** illustrates the lower post end **214** of the reinforcement post **210**. As shown, the strap engagement slot **222** can extend substantially vertically upward into the reinforcement post **210** at the lower post end **214**. In example aspects, the strap engagement slot **222** can be defined by a first slot **1510** formed in the first post side **312** of the reinforcement post **210** and a second slot (not shown) formed in the second post side **314** (shown in FIG. **3**) of the reinforcement post **210** that can be laterally aligned with the first slot **1510**. The second slot can be substantially the same in size and shape as the first slot **1510**. The strap engagement slot **222** can extend laterally across the reinforcement post **210** from the first post side **312** to the second post side **314**. In some aspects, the strap engagement slot **222** can be positioned closer to the rear post side **320** than to the front post side **218**. In other aspects, however, the strap engagement slot **222** can be arranged at any suitable position between the rear post side **320** and the front post side **218**. According to example aspects, the lower guardrail strap **125***c* (shown in FIG. **1**) can extend laterally through the strap engagement slot **222** to secure the lower guardrail strap **125***c* to the reinforcement post **210**. In other aspects, the lower guardrail strap **125***c* can be secured to the reinforcement post **210** by a lower one of the guardrail clamp assemblies **230** (shown in FIG. **2**). Other aspects may not comprise any of the lower guardrail strap **125***c*, the strap engagement slot **222**, and the lower one of the guardrail clamp assemblies **230**.

[0068] FIG. **16** illustrates a cross-sectional view of the upper clamp assembly **230***a* coupled to the reinforcement post **210**, in accordance with another example aspect of the present disclosure, taken along a line similar to line **6-6** in FIG. **5**. The upper clamp assembly **230***a* and the reinforcement post **210** can be substantially similar to the upper clamp assembly **230***a* and the reinforcement post **210** previously described. However, in the present aspect, the reinforcement post assembly **200** does not comprise the guide rivet nut **640** (shown in FIG. **6**) mounted within the front post opening **636** at the front post side **218** of the reinforcement post **210**. According to example aspects, upon tightening the primary slot stop **360** to extend the threaded shaft **336** thereof through the front post opening **636**, a clearance can be defined between the threaded shaft **336** of the primary slot stop **360** and the front post opening **636**, thereby allowing the threaded shaft **336** to move easily within the front post opening **636**. Furthermore, in some aspects, the threaded shaft **336** of the primary slot stop **360** can be sized such that the distal stop end **362** of the threaded shaft **336** can be spaced from the rear post side **320** of the reinforcement post **210** in the blocked position. In other aspects, the distal stop end **362** may contact the rear post side **320**. Other aspects of the clamp assembly **230** may not comprise the front post opening **636**, and the distal stop end **362** of the threaded shaft **336** can abut the front post side **218** in the blocked position.

[0069] One should note that conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular embodiments or that one or more particular embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

[0070] It should be emphasized that the above-described embodiments are merely possible examples of implementations, merely set forth for a clear understanding of the principles of the present disclosure. Any process descriptions or blocks in flow diagrams should be understood as representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process, and alternate implementations are included in which functions may not be included or executed at all, may be executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved, as would be understood by those reasonably skilled in the art of the present disclosure. Many variations and modifications may be made to the above-described embodiment(s) without departing substantially from the spirit and principles of the present disclosure. Further, the scope of the present disclosure is intended to cover any and all combinations and sub-combinations of all elements, features, and aspects discussed above. All such modifications and variations are intended to be included herein within the scope of the present disclosure, and all possible claims to individual aspects or combinations of elements or steps are intended to be supported by the present disclosure.

Claims

1. A clamp assembly for a fall protection reinforcement post, the clamp assembly comprising: a clamp body at least partially defining a slot configured to receive a guardrail therethrough; a post fastener engaging the clamp body and configured to secure the clamp assembly to the fall protection reinforcement post; and a clamp spring biasing the clamp assembly to an untightened configuration.
2. The clamp assembly of claim 1, wherein: the clamp body defines: a front clamp wall configured to confront a front post side of the reinforcement post; and a rear clamp wall opposite the front clamp wall and configured to confront a rear post side of the reinforcement post; and the slot is at least partially defined by the front clamp wall.
3. The clamp assembly of claim 2, wherein the post fastener extends through a rear clamp opening in the rear clamp wall towards the front clamp wall and is configured to engage the reinforcement post at the rear post side.
4. The clamp assembly of claim 3, wherein the post fastener is a threaded bolt fastener comprising a threaded shaft and an adjustment knob, the threaded shaft extending through the rear clamp opening.
5. The clamp assembly of claim 2, wherein the clamp spring is arranged within the slot and engages the front clamp wall, the clamp spring configured to bias the front clamp wall forwardly away from the front post side of the reinforcement post in the untightened configuration.
6. The clamp assembly of claim 5, wherein the clamp spring is a button spring clip comprising a substantially inverted V-shaped resilient spring portion and a button tab extending from the substantially inverted V-shaped resilient spring portion.
7. The clamp assembly of claim 6, wherein: the substantially inverted V-shaped resilient spring portion defines a first leg and a second leg; the first leg confronts an inner clamp wall surface of the front clamp wall; the second leg is configured to confront an outer post surface of the front post side; and the button tab extends from the second leg and is configured to engage a button opening formed through the front post side.
8. The clamp assembly of claim 2, wherein: the clamp body further defines: a first clamp sidewall extending between the front clamp wall and the rear clamp wall; and a second clamp sidewall extending between the front clamp wall and the rear clamp wall, opposite the first clamp sidewall; each of the first clamp sidewall and the second clamp sidewall define a notch; and a forward edge of each notch at least partially defines the slot.
9. The clamp assembly of claim 8, wherein a plurality of gripping teeth are formed along the forward edge of each notch.
10. The clamp assembly of claim 8, wherein at least one of the first clamp sidewall or the second clamp sidewall defines a slot stop extending towards the rear clamp wall and at least partially traversing the slot.
11. The clamp assembly of claim 10, wherein the slot stop is formed as a stop hook at a bottom wall end of the first clamp sidewall or the second clamp sidewall.
12. The clamp assembly of claim 1, further comprising a clamp end cap mounted to the clamp body at an upper clamp end of the clamp body.
13. A reinforcement post assembly comprising: a fall protection reinforcement post defining a front

post side and a rear post side; and a clamp assembly comprising: a clamp body; a post fastener engaging the clamp body and the reinforcement post to secure the clamp assembly to the reinforcement post; and a clamp spring biasing the clamp assembly to an untightened configuration; and wherein a slot is defined between the clamp body and the front post side of the reinforcement post, the slot configured to receive a guardrail therethrough.

14. The reinforcement post assembly of claim 13, wherein: the clamp body defines a front clamp wall confronting the front post side and a rear clamp wall opposite the front clamp wall and confronting the rear post side; the clamp spring is arranged within the slot between the front clamp wall and the front post side; and the clamp spring is a button spring clip comprising a substantially inverted V-shaped resilient spring portion and a button tab extending from the substantially inverted V-shaped resilient spring portion.

15. The reinforcement post assembly of claim 14, wherein: the substantially inverted V-shaped resilient spring portion defines a first leg and a second leg; the first leg confronts an inner clamp wall surface of the front clamp wall; the second leg confronts an outer post surface of the front post side; and the button tab extends from the second leg and engages a button opening formed through the front post side.

16. The reinforcement post assembly of claim 13, wherein: the clamp body defines: a front clamp wall confronting the front post side; a rear clamp wall opposite the front clamp wall and confronting the rear post side; a first clamp sidewall extending between the front clamp wall and the rear clamp wall; and a second clamp sidewall extending between the front clamp wall and the rear clamp wall, opposite the first clamp sidewall; each of the first clamp sidewall and the second clamp sidewall defines a notch; and a forward edge of each notch at least partially defines the slot.

17. The reinforcement post assembly of claim 16, wherein a plurality of gripping teeth are formed along the forward edge of each notch.

18. The reinforcement post assembly of claim 16, wherein at least one of the first clamp sidewall or the second clamp sidewall defines a slot stop extending towards the rear clamp wall and at least partially traversing the slot.

19. The clamp assembly of claim 18, wherein the slot stop is formed as a stop hook at a bottom wall end of the first clamp sidewall or the second clamp sidewall.

20. The clamp assembly of claim 19, further comprising a clamp end cap mounted to the clamp body at an upper clamp end of the clamp body.
